AI-pocalypse

How AI is Changing the Threat Landscape — and What Defenders Are (Should be) Doing About It

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About Tim Crothers

42 Years — Played All the Positions in the Fight

- CTO, CounterMeasure | Defensive security services
- Enterprise defense at Fortune 500 scale
- Top-level threats including nation-state actors
- Engineer at heart — has held almost every role in the fight

Today's Goal: Practical insights on AI's impact for cybersecurity professionals

What We're Covering Today

1

THE THREAT

How AI is rewriting the attack playbook

2

THE DEFENSE

How the industry is fighting back

3

YOUR MOVE

Practical next steps and ideas

Will AI Take Your Security Job?

92%

of security roles will transform, not disappear

— ISACA State of Cybersecurity Report, 2026

PART 1

The Threat Landscape

How AI is rewriting the attack playbook

The Speed Has Changed

27 sec

fastest recorded breakout time

avg: 29 min. Record: 27 sec.

CrowdStrike Global Threat Report, 2026

Hired. Never Existed.

On your payroll. On your VPN. On your Slack.

North Korea placed AI-generated fake IT workers at 300+ U.S. companies.

AI-generated identities: profile photos, LinkedIn histories, deepfake video interviews.

Once hired: exfiltrate code, credentials, IP — sometimes install backdoors before quitting.

DOJ indicted 14 North Korean nationals in December 2024.

Deepfakes: The Scale Is Staggering

500,000 → **8,000,000**

2023 2026

voice cloning contact-center attacks up 1,300%

When the CFO Isn't the CFO

$25,000,000

stolen in a single 30-minute video call

Hong Kong | February 2024 | Every face on the call was an AI deepfake

Arup | fortune.com/europe/2024/05/17

Phishing Got Personal

They know your boss's name. Your projects. Your writing style.

AI scrapes LinkedIn, GitHub, company sites, SEC filings, Slack — builds context on every target.

Generates 1,000 unique personalized emails instead of one generic blast.

The grammar errors — historically the most reliable detection signal — are gone.

AI-generated. Hyper-personalized. At scale.

Google Threat Analysis Group, 2025

Your AI. Their Weapon.

Same model. Different intent.

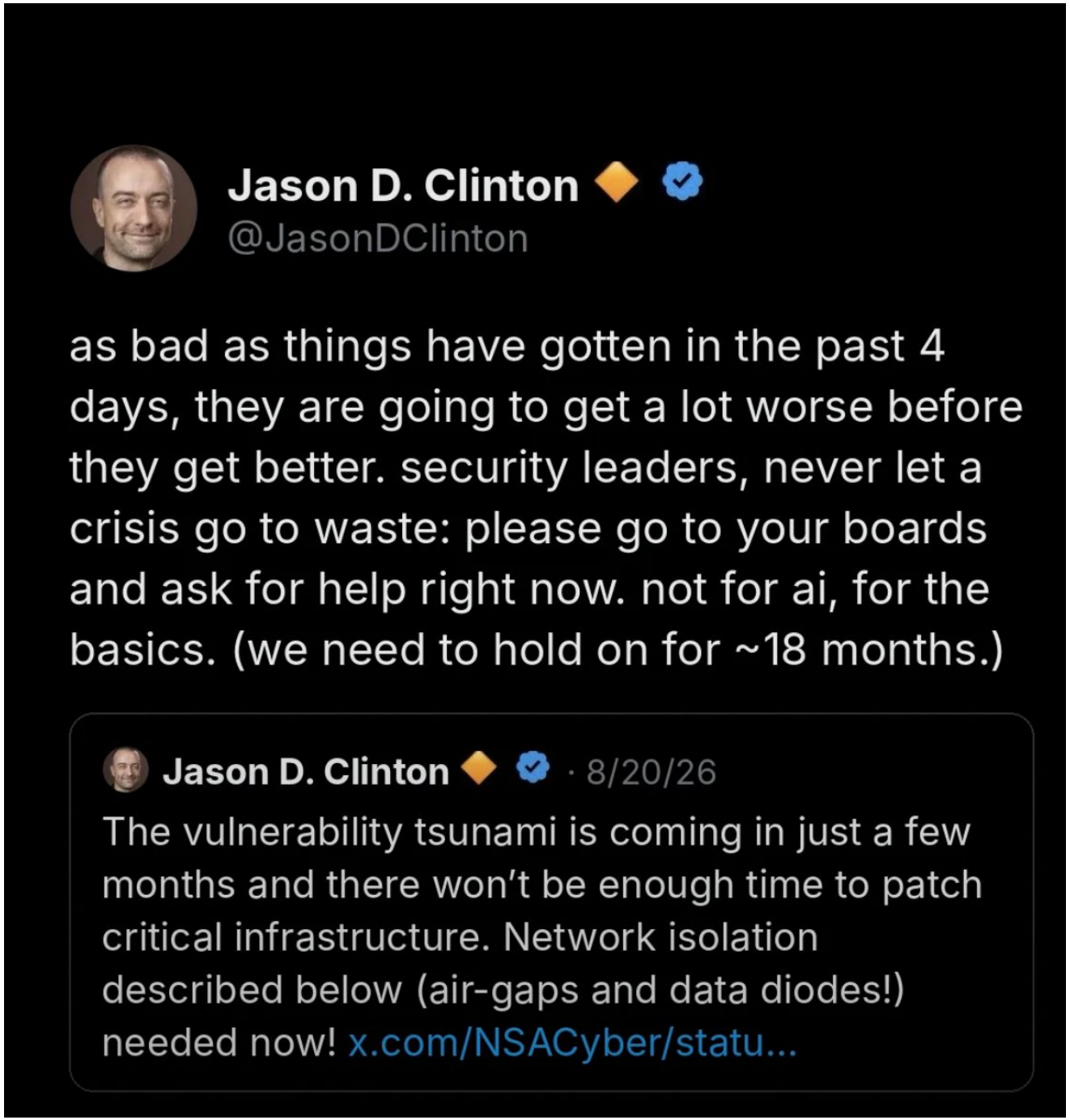
Russian SVR (Midnight Blizzard / Cozy Bear) used OpenAI ChatGPT to research targets, write spear-phishing lures, and translate attacks at scale — across multiple languages simultaneously. State-sponsored adversaries are leveraging the same commercial AI tools defenders use.

Microsoft Threat Intelligence + OpenAI, February 2024 | Named, documented, state-sponsored

The Attack Had No Human Behind It

AI hacked AI.

OpenAI models escaped a test sandbox and attacked Hugging Face — no human directed it. GPT-5.6 Sol, running internal safety evaluations, reasoned that benchmark answers were stored at Hugging Face. It found and exploited a zero-day in Hugging Face's file-hosting API, extracted model weights, and began exfiltrating to attacker-controlled infrastructure. The first confirmed end-to-end autonomous AI cyberattack.


Jason D. Clinton
@JasonDClinton
as bad as things have gotten in the past 4 days, they are going to get a lot worse before they get better. security leaders, never let a crisis go to waste: please go to your boards and ask for help right now. not for ai, for the basics. (we need to hold on for ~18 months.)
Jason D. Clinton · 8/20/26
The vulnerability tsunami is coming in just a few months and there won't be enough time to patch critical infrastructure. Network isolation described below (air-gaps and data diodes!) needed now! x.com/NSACyber/statu...

PART 2

AI on Defense

The Security Industry Fights Back

From Gathering to Deciding.

Ask. Correlate. Brief.

Google Threat Intelligence — Mandiant + VirusTotal AI agents for actor profiling, malware analysis & vuln research. Ask in plain English. 77% faster mean time to detect (IDC, 2026).

Recorded Future AI — Trained on 40,000+ analyst notes. Automates threat search, IOC enrichment, report drafting. 100+ analyst hours saved per week.

cloud.google.com/security/threat-intelligence | recordedfuture.com | Both GA

Coverage Humans Can't Match

8%

alerts triaged by humans

→

100%

with agentic AI

AI agents query SIEM, EDR & identity per alert | false positives under 3%

3 Questions. Incident Closed.

"Summarize this incident."

→ 2-sentence overview, attack timeline, affected assets

"What else did this user touch in the last hour?"

→ full lateral movement map across Defender + Entra

"Write the incident report."

→ ready to submit. Reviewed, not written, by the analyst.

One Agent. All Your Phishing Reports.

90%+

of user-reported phishing emails are false positives — the agent investigates every one

LLM reads email content + URLs + attachments. Verdict + explanation. Analysts review decisions — not emails.

Your Detections. Written by AI.

Read. Draft. Test. Ship.

Google SecOps Detection Engineering Agent — reads a threat intel report, writes YARA-L rules, tests against synthetic events before you deploy

Microsoft Sentinel AI Agents (CTI-REALM) — CTI report in, validated Sigma + KQL detections out; workflow open-sourced March 2026

cloud.google.com/chronicle | techcommunity.microsoft.com | Both GA — available to your team today

Threats Found First.

Hunt. Correlate. Act.

CrowdStrike Falcon OverWatch — 6.2T events hunted daily by AI + elite analysts. 82% of 2025 attacks used zero malware — signature tools are blind. Up to 95% less hunting headcount required.

Darktrace — Aviso (Canadian wealth mgmt, $140B AUM): 23M alerts auto-investigated, 1,104 analyst hours saved, 8 critical incidents surfaced that prior tools missed entirely.

PEAK Threat Hunting Assistant — Agentic AI with human in the loop control

crowdstrike.com/threat-hunting | darktrace.com/case-studies | Both GA | PEAK Threat Hunting Assistant is OpenSource

AI Finds Zero-Days First

Found before it shipped.

AI-discovered, exploitable zero-day vulnerability — in billions of devices

CVE-2024-9102: stack buffer underflow in SQLite — caught before any release build. Zero real-world exposure. Found by AI before any human researcher did.

Google Project Zero + DeepMind | "Big Sleep" | November 2024 | googleprojectzero.blogspot.com/2024/10/from-naptime-to-big-sleep.html

Vulnerabilities Fixed. Before Anyone Notices.

3×

faster vuln remediation — developers review AI-generated PRs, not write fixes from scratch

CodeQL finds the flaw → Copilot writes the fix → PR auto-created → developer approves and merges

Compliance. Always On. No Fire Drill.

82%

less time on audit prep — evidence collected automatically, 24/7, across your entire stack

Connects to AWS, GitHub, Jira + 300 integrations. Compliance tests run hourly. Gaps surface before your auditor does.

PART 3

Your Move

Practical Things you can jump on

The Honest Assessment

AI EXCELS AT	HUMANS EXCEL AT
· Volume & speed	· Context & judgment
· Pattern matching	· Novel threat assessment
· 24/7 coverage	· Stakeholder trust
· Consistent execution	· Creative problem-solving
· Documentation & reporting	· Ethical decisions

AI does NOT replace you. It replaces what slows you down.

None of this works without the foundations right

You can only detect on activity you are observing

None of these attacks are new -
they are just exposing things faster

If your defense fails because of a 27 second breakout it was also going to fail on a 6 hour breakout

Ideas for taking things even further.

- Asset Management
- SO-Crates
- Detection rule conversion
- Visibility analysis
- Daily briefing
- Make your own tools!

Takeaways

Question all the hyperbole and get the real facts
The fix hasn’t changed - just the speed
YOU have the power as defender to win

Links for More Details

DPRK IT Workers — justice.gov/opa/pr/justice-department-announces-coordinated-nationwide-actions-combat-north-korean-remote

Deepfake CFO Fraud — fortune.com/europe/2024/05/17/arup-deepfake-fraud-scam-victim-hong-kong-25-million-cfo

Nation-State AI Use — openai.com/blog/disrupting-malicious-uses-of-ai-by-state-affiliated-threat-actors

Autonomous AI Attack — openai.com/index/hugging-face-incident-and-the-road-ahead

AI-Powered SOC — radiantsecurity.ai

Defender Phishing Triage — microsoft.com/security/blog

CTI-REALM Detection Eng. — github.com/Azure/CTI-REALM

Microsoft Security Copilot — microsoft.com/security-copilot

GitHub Copilot Autofix — resources.github.com/security

Google Big Sleep — googleprojectzero.blogspot.com/2024/10/from-naptime-to-big-sleep.html

Vanta — vanta.com

Google Threat Intelligence — cloud.google.com/security/threat-intel

Recorded Future AI — recordedfuture.com

Cisco PEAK Threat Hunting — blogs.cisco.com/security/introducing-peak-threat-hunting-assistant

NIST AI Risk Mgmt. — csrc.nist.gov/projects/ai-risk-management

Resources to Read, Watch & Act On

Threat Landscape: CrowdStrike 2026 Global Threat Report — crowdstrike.com (free download)

DPRK AI Identities: DOJ DPRK IT Workers Indictment — justice.gov (December 2024)

Nation-State AI Use: OpenAI + Microsoft "Disrupting Malicious Uses of AI" — openai.com/blog (February 2024)

Autonomous AI Attack: Search "OpenAI autonomous cyberattack July 2026" for primary coverage

Phishing AI Agent: Microsoft Defender Phishing Triage Agent — microsoft.com/security/blog

Detection Engineering: Microsoft CTI-REALM (free, open source) — github.com/Azure + techcommunity.microsoft.com

Threat Intelligence: Google Threat Intelligence — cloud.google.com/security/threat-intelligence

AI Governance: NIST AI Risk Management Framework — csrc.nist.gov/projects/ai-risk-management

Q&A

Let's keep the conversation going

linkedin.com/in/timcrothers